POWER BI
PROJECT REPORT

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PROJECT TITLE :	**NETFLIX CONTENT ANALYSIS**
PROJECT DURATION :	**3 DAYS**
TRAINER NAME :	**MR. SAYAN SAU**
SUBMISSION DATE :	**JULY 11, 2026**
GITHUB USERNAME :	**aakashnath**
GITHUB REPOSITORY LINK :	**https://github.com/aakashnath/personal-finance-dashboard-excel**

1. PROJECT OVERVIEW

The **Netflix Content Analysis** is an advanced Power BI project developed to analyze Netflix's global content catalogue and present the findings through an interactive, visually engaging dashboard. The project transforms raw, unstructured data into clear and meaningful visual insights, allowing viewers to explore Netflix's content library across genres, ratings, countries, and release years.

Power BI was chosen for this project because of its strong data modelling capabilities, intuitive drag-and-drop visual design, and ability to build interactive, filterable dashboards without extensive coding. It allows large datasets to be cleaned, transformed, and visualized within a single environment using **Power Query** and **DAX**, making it an ideal tool for business intelligence and analytics projects of this scale.

Interactive dashboards offer significant advantages over static reports. They allow end users to filter, drill down, and explore data on their own terms, uncovering patterns and relationships that a fixed report would not reveal. For a streaming platform such as Netflix, this kind of analysis helps business users understand what type of content performs well, which regions contribute the most titles, and how the content library has evolved over time — insights that directly support content acquisition, marketing, and regional strategy decisions.

This project uses a publicly available **Netflix Movies & TV Shows** Dataset downloaded from **Kaggle.** Since the dataset was published approximately five years ago, the available data covers Netflix content released between **1925** and **2021.** The objective of this project is to transform raw Netflix data into an interactive Power BI dashboard that supports data-driven decision-making and business intelligence.

2. PROJECT OBJECTIVES

The main objectives of this project are:

- Analyze Netflix Movies and TV Shows.
- Identify content distribution by country.
- Analyze content ratings.
- Compare Movies vs TV Shows.
- Analyze popular genres.
- Study release year trends.
- Build an interactive dashboard.
- Generate meaningful business insights.

3. DATASET DESCRIPTION

Dataset Name: Netflix Movies & TV Shows Dataset

Source: Kaggle (Public Dataset)

The dataset used in this project was downloaded from Kaggle by searching for **"Netflix Movies and TV Shows."** The downloaded file was provided in a compressed ZIP format, which was extracted to access the underlying CSV file. Before importing the data into Power BI, the dataset was first opened in Microsoft Excel for an initial inspection, allowing a quick review of the column structure, data types, and overall quality of the records. Once verified, the dataset was imported into **Power BI** Desktop for further cleaning, transformation, and visualization.

The dataset contains over **8,000** records covering both **Movies and TV Shows** available on Netflix. Since the dataset was published approximately five years ago, it spans Netflix content released between **1925** and **2021**, which is why the dashboard's release-year analysis ends at **2021** rather than reflecting the current year.

The dataset includes several important fields used throughout the analysis, including **Title, Type, Director, Cast, Country, Date Added, Release Year, Rating, Duration, Genre (renamed from listed_in for better readability), and Description**. The original dataset remained unchanged, with only selected field names being renamed in Power BI to improve clarity and presentation.

4. DATA CLEANING & PREPROCESSING

Before visualization, the dataset was cleaned and prepared using the following steps:

- Extracted the downloaded ZIP dataset to access the raw CSV file.
- Opened the dataset in Microsoft Excel for initial verification of structure and content.
- Imported the dataset into Power BI Desktop for further processing.
- Removed duplicate records to ensure data accuracy.
- Checked and handled missing or null values in key fields.
- Corrected inconsistent data types across columns.
- Standardized country names where required for uniform analysis.
- Verified the Release Year, Rating, and Duration fields for consistency.
- Used Power Query for data cleaning and transformation.
- Prepared the final dataset for dashboard visualization.

5. TOOLS & TECHNOLOGIES USED

The following tools and technologies were used to build this project:

- Microsoft Excel (Initial Dataset Verification)
- Power BI Desktop
- Power Query
- DAX
- Data Cleaning
- Data Transformation
- KPI Cards
- Bar Charts
- Filled Map
- Donut Chart
- Area Chart
- Interactive Dashboard

6. DASHBOARD FEATURES

KPI Cards: Display key summary metrics at a glance, including total movies, total TV shows, total genres, and the release-year range covered by the dataset.

Rating by Movies: A horizontal bar chart comparing the number of titles across content ratings such as TV-MA, TV-14, and PG-13, highlighting audience-suitability trends.

Genres by Total Shows: A bar chart ranking genres by the number of titles, revealing which categories such as Drama and Documentaries dominate the catalogue.

Count of Shows by Country: A filled map visualization showing the geographic distribution of Netflix titles, making it easy to identify the leading content-producing regions.

Movies vs TV Shows: A donut chart comparing the overall proportion of movies to TV shows within the catalogue.

Total Movies by Release Year: An area chart tracking the number of titles released each year, illustrating Netflix's content growth over time.

7. DASHBOARD SCREENSHOT

Fig. 1: Netflix Content Analysis Dashboard

8. KEY INSIGHTS

- Movies dominate Netflix's overall catalogue.
- TV-MA is the most common content rating.
- Drama is the leading content genre.
- The United States has the highest number of titles.
- Netflix content increased significantly after 2015.
- The dashboard enables quick exploration of global content trends.

9. BUSINESS VALUE

The **Netflix Content Analysis Dashboard** delivers meaningful business value across several areas of decision-making. It supports content strategy by highlighting which genres and ratings perform best within the catalogue, helping stakeholders identify opportunities for future content acquisition. It aids regional planning by revealing which countries contribute the highest number of titles, allowing for more informed decisions around regional content investment.

The dashboard also supports audience analysis by showing how ratings and genres are distributed, offering insight into viewer preferences. For executive decision-making, the KPI cards and visual summaries provide a quick, high-level overview that enables leadership to assess the catalogue at a glance. Additionally, the release-year trend visualization assists in trend identification, showing how content volume has evolved over time, while the overall dashboard functions as a reliable tool for business intelligence reporting.

10. LEARNING OUTCOMES

Through this project, the following practical skills were developed:

- Power BI Dashboard Development
- Data Cleaning
- Power Query
- DAX Basics
- Data Visualization
- KPI Design
- Interactive Dashboard Design
- Business Analytics

11. CONCLUSION

The **Netflix Content Analysis Dashboard** demonstrates how Power BI can be effectively used to simplify complex content data into clear, actionable insights. By combining structured data cleaning with well-designed KPI cards, charts, and maps, the dashboard provides meaningful business insights into Netflix's content library, including genre popularity, rating distribution, and regional content trends.

This project successfully demonstrates practical **Power BI** skills, including data cleaning with Power Query, dashboard design, and interactive visualization, while supporting data-driven decision-making for business stakeholders. Going forward, this project can be further enhanced through real-time data integration and more advanced analytics, such as predictive modelling of content trends using DAX or machine learning tools.

TRAINER VERIFICATION SECTION

Trainer Remarks: ____

Signature: ____

Date: ___________ / _____________ / _____________